Retrospective reconstruction and numerical qualification of historical countermovement-jump recovery research: An evidence-bounded case study

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Short title: A case study in jump recovery research

Abstract

This retrospective case study asks how partially preserved historical machine-learning research can be reconstructed while distinguishing source-equivalent computation, compatibility with archived outputs, measurement authority, public reproducibility, and predictive validity. The archive supports eight historical scientific formulations and nine neural model families. Public records identify their structures; original route qualification remains a privately recorded result. Those records report exact equality between the recovered reference implementation and original central processing unit (CPU) outputs for all 18 routes. Comparisons with archived graphics processing unit (GPU) arrays pass for M0–M3 and fail for both repeats of M4–M8 under the unchanged tolerance. This is not a public rerun of the original checkpoints or a successful replay of the historical GPU environment. Separately, we report an existing public synthetic benchmark with four reference models and three fixed evaluation roots from one generator. Mean six-cell normalized root-mean-square error (NRMSE) was 2.430659 for zero innovation, 1.036038

for the training-mean baseline, 0.736270 for ridge regression, and 0.773316 for the structured comparator. These descriptive synthetic results do not estimate predictive performance of the historical model families. Neither the synthetic benchmark nor numerical parity establishes athlete predictive validity, transportability, or clinical utility. No new training, inference, resampling, athlete-data acquisition, or empirical scoring was performed for this article.

Keywords: scientific machine learning; reproducible research; model reconstruction; countermovement jump; synthetic benchmark; numerical reproducibility

1 Introduction

Reconstructing a scientific machine-learning result requires more than locating executable source code. The result also depends on fitted state, feature ordering, preprocessing, target definition, split membership, runtime behavior, and aggregation. When these pieces are preserved separately or incompletely, a successful present-day run can answer a different question from the historical experiment. Goodman and colleagues describe reproducibility terminology as unsettled across fields; we therefore report the operation and evidence for each result instead of assigning one binary label [1]. The NeurIPS reproducibility program described by Pineau and colleagues combined code policy, a community challenge, and a submission checklist to improve prospective reporting and evaluation [2]. The present work applies those concerns retrospectively to one archive and does not claim that checklists or existing research infrastructure are absent [3].

Several established systems support prospective experiment capture and artifact identity. MLflow Tracking documents APIs for recording parameters, code versions, metrics, and output artifacts for runs; DVC documents experiment management alongside data, model, and pipeline versioning [4, 5]. Provenance and research-object standards address related needs: W3C PROV-O supplies a vocabulary for provenance entities and activities, while RO-Crate specifies packaging of research objects and their metadata [6, 7]. FAIR4RS describes findability, accessibility, interoperability, and

reusability goals for research software [8]. These sources are complementary prior work, not systems benchmarked in this study. We do not claim that they cannot represent historical records, restricted artifacts, or provenance. Our narrower task is to recover what this particular archive supports and label what remains incomplete, private, or unresolved; the materials accompanying this article do not claim conformance to PROV-O or RO-Crate.

CMJ measurement contracts make the distinction between software and science concrete. Force-time outcomes depend on event boundaries and measurement procedures; they do not make force and impulse interchangeable outcomes [9, 10]. The historical and public synthetic tasks also have different horizons, targets, and evidence authority. The case-specific question is what can be authenticated from the preserved records without treating computational agreement as evidence of measurement validity or athlete prediction.

The case study distinguishes eight historical scientific formulations from nine original neural model families, identified in the source archive as M0–M8 and each represented by two repeats (r0 and r1). A separate, previously recorded public synthetic study compares four reference models across three horizons and two outcomes. Their target definitions, result authority, and reproducibility evidence differ.

RQ1 — Historical reconstructability. Which parts of the original research program can be authenticated and reconstructed, and which remain partial, rights-restricted, or unresolved?

RQ2 — Numerical reproducibility. Does recovering the original computation and obtaining source-equivalent CPU inference establish compatibility with saved historical GPU outputs?

RQ3 — Scientific comparability. What conclusions are supported by the public fixed-mode synthetic benchmark, and why do they not establish validity of the original native recovery models in athletes?

The contribution is a retrospective case study, not a new universal provenance framework, neural-network algorithm, or validated athlete predictor. It shows how

this archive's formulation identities, privately recorded route results, public synthetic-study results, and measurement contracts lead to different evidence judgments. An approach that binds claims to source identity and reports public reproducibility separately from restricted qualification may be useful in other incomplete archives, but transferability has not been experimentally tested here.

2 Related work

2.1 Research reproducibility and machine-learning reporting

The NeurIPS reproducibility program reported by Pineau et al. combined a code submission policy, a community-wide reproducibility challenge, and a paper checklist [2]. Goodman et al. explain why disciplines use reproducibility and related terms inconsistently and caution against using them as proxies for truth [1]. We use operational descriptions here: public regeneration refers to calculations reproducible from the public data package; private qualification refers to recorded checks whose underlying artifacts are not distributed; and scientific validity refers to evidence about the intended measurements and population. These labels prevent a recorded model execution from being presented as a public rerun or a predictive-validity study.

2.2 Experiment tracking, versioning, and provenance standards

MLflow Tracking documents run-level logging for parameters, code versions, metrics, and artifacts. DVC documents experiment management and versioning workflows for data and models [4, 5]. The PROV Ontology provides a standard vocabulary for representing provenance; RO-Crate defines a packaging specification for research objects, metadata, data, and workflows [6, 7]. FAIR4RS adapts the FAIR principles to research software [8, 11]. These infrastructures address adjacent problems and can be used to record historical material when it is available. We neither evaluate their capabilities nor claim that they cannot support retrospective recovery. The present study uses source files from the published software release, SHA-256 checksums, and claim-to-evidence

records for this case; it does not claim conformance to PROV-O or RO-Crate.

2.3 Artifact review and numerical compatibility

ACM's Artifact Review and Badging policy v1.1 separates artifact evaluation, public artifact availability, and results validation; functional and reusable artifacts are distinct evaluation levels [12]. The policy permits artifact evaluation to be separate from public availability, while its reproduced and replicated result labels distinguish whether author-supplied artifacts are used. We use a related separation of evidence questions, but this manuscript has not undergone ACM artifact review and carries no badge. Its private original-model checks are recorded qualification, not an independent public artifact evaluation. Floating-point literature shows why numerical comparison must be tied to a stated operation and tolerance; it does not identify the cause of the discrepancies in this archive [13,14].

2.4 Measurement and simulation-study reporting

CMJ reporting literature emphasizes that selected force-time metrics and their measurement procedures should be explicit [9,10]. TRIPOD-Cluster motivates transparent descriptions of clustered development and validation data [15], but its reporting guidance does not make synthetic camps equivalent to human cohorts. Morris, White, and Crowther propose the ADEMP structure—Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures—for planning and reporting simulation studies [16]. We use ADEMP to describe the existing public synthetic experiment, not to claim a new simulation or to generalize beyond its generator. The ML test-score rubric concerns engineering practices and production readiness [17]; we cite it only to distinguish those questions from scientific validation and assign no readiness score.

3 Materials and methods

3.1 Study design and evidence scope

This work is a retrospective reconstruction and evidence synthesis. The scientific software identity is the published v1.1.0 release at commit 401f87aad1e4ef4108011fa0c1d769a24f1ad329. The public synthetic-study results retain their original source and package identities; they are not attributed to the software release commit. Source-file hashes, the claim-to-evidence register, and numerical summaries accompany the article.

Model inventories, configuration links, checkpoint selection, fitted-transform digests, and numerical results were reconciled during private qualification. The public materials contain only summaries needed to describe those checks. Original checkpoints, fitted parameters, and restricted scientific records are not included. The route-level matrix therefore presents privately qualified evidence as a recorded result; it does not let a reader rerun inference against the private arrays.

The manuscript preparation performed no new training, model fitting, checkpoint inference, or empirical scoring. Figures use the fixed public results, stored study summaries, public route records, and recorded methods. The prior qualification independently checked 306 RMSE-to-standard-deviation ratios and 51 root/cell aggregate identities. The six bootstrap summaries were retained rather than recomputed. Any additional arithmetic in this manuscript is limited to presentation of those recorded aggregates.

3.2 Historical scientific formulations

The archive records eight formulation identities with different response worlds, observation settings, horizons, and dispositions. A direct parent is shown only when it is explicitly present in the lineage record. The native camp formulation forecasts 72-hour (H72) and day-7 (D7) targets. The initial and canonical preseason formulations share an event-time adaptation world; canonicalization changes the sample and query sur-

Fig 1. Scientific formulation genealogy. Cards show the eight recorded formulations with their disposition, response world, observation contract and forecast horizons (dots). Arrows are direct parent relationships recorded in the published lineage and methods records: initial preseason to canonical preseason; preliminary episode to phase-consistent episode to correlated exposure; and correlated exposure to the threshold and fixed-mode alternatives. Dashed stubs marked ? show the three identities whose direct scientific parent is not recorded (initial preseason, rich-history camp and preliminary episode). The diagram describes the archive, not a single linear history, a calendar timeline or an accepted physiological theory.

Fig 2. Structure of the existing public fixed-mode synthetic study. (a) Each square is one camp (250 episodes, 750 horizon rows): 96 training camps, 16 development camps, and three fixed evaluation roots of 16 camps each. (b) Horizon-row counts per split, with episode counts (training 24,000; development and each root 4,000; three rows per episode). (c) Each episode contributes H24, H48 and H72 queries that remain in one split. Participant, episode, query and camp identities are checked for split separation; group keys, labels, future target-time fields and hidden or oracle states are excluded from predictors. The roots share one generator and are not human cohorts.

face. Rich-history camp recovery is a separate retired formulation with unresolved ancestry. The episode formulations start with a preliminary paired-trial observation contract, then a phase-consistent force-impulse measurement, followed by a correlated-exposure world. These episode tasks query 24-hour (H24), 48-hour (H48), and 72-hour (H72) horizons. Threshold response is a proposed alternate. The fixed-mode discrepancy formulation is an active candidate with mathematical 24-hour and 84-hour response modes; those time constants are not identified biological compartments.

Fig. 1 shows the recorded links and leaves unresolved parents open. The native camp task uses H72 and D7 targets. The episode formulations and the public synthetic benchmark use H24, H48, and H72 horizons. Shared outcome labels do not make their measurement laws, generators, or score definitions interchangeable.

Fig. 2 reports the synthetic-study split sizes; the three fixed evaluation roots share one generator, and each episode's H24, H48, and H72 queries remain grouped.

3.3 Original neural model reconstruction

The historical model inventory contains nine architecture families identified as M0–M8, with repeat identifiers r0 and r1. These original identifiers are retained to connect each result to the qualification record. The recovered input contract contains 39 standardized raw predictors. M3 and later families also use four history records with four values each. M5 adds a two-entry horizon embedding with eight dimensions per entry.

Fig 3. Historical neural architectures and complexity. (a) Computational graph of the largest family, M8, with each module labelled by the first family that contains it. M0–M2 use only the raw feedforward backbone; M3 adds a pointwise history encoder and M4 replaces it with a 32-unit GRU, both followed by mean and latest-value pooling; M5 adds a two-entry, eight-dimensional horizon embedding; M6 projects the first two of five schedule-basis values and M7 all five; M8 adds a bias-free 5-to-2 linear basis head. The concatenated width is 128 (M3, M4), 136 (M5) or 152 (M6–M8). (b) Component inclusion and recorded trainable parameter counts (1,346 to 10,796). Repeats r0 and r1 share each family's topology and count. Counts are from the public model-qualification summary. The figure describes structure, not predictive accuracy; fitted weights are not shown.

M6–M8 receive five schedule-derived basis values. The basis uses exposure durations normalized by 100 minutes, exponential time weights of 36 and 120 hours, and a saturating cumulative-dose transform $h(C) = 10C/(1 + C)$. It is a schedule feature, not a measured physiological state.

Original fitted preprocessing values were restored from authenticated configuration records in the private qualification process; they were not refit. Standardization preceded conversion to the neural network input type. The model output is an additive innovation for force followed by net impulse, with an identity output transform. Adding the baseline is outside the inference adapter. The fitted parameter values and private checkpoint hashes are not published.

The five-value original schedule basis requires complete history and future-plan snapshots that were genuinely known at the forecast origin. The index exposure is included once in the known plan, not added a second time. The basis function reconstructs schedule features but cannot authenticate when an empirical future plan became available. A predetermined synthetic plan is not evidence of prospective availability in an athlete archive.

Table 1 summarizes the published architecture record and parameter counts for M0–M8. Both repeats share each family's listed architecture.

Fig. 3 shows the corresponding branches: M0–M2 are raw-feature feedforward networks of increasing width and depth; M3 encodes pointwise history and pools its mean and latest values; M4 substitutes a 32-unit gated recurrent unit (GRU); and M5 adds a horizon embedding. M6 projects the first two basis components, M7 projects all five, and M8 adds a bias-free five-to-two linear basis head to its residual prediction.

Table 1. Architecture contracts and parameter counts for nine historical neural model families (M0–M8). Each family has two recorded repeats (r0 and r1). Input counts and architecture descriptions are from the public model-qualification summary; original checkpoint weights are not distributed.

Family	Additional inputs	Recorded architecture	Parameters
M0	None	$39 \to 32 \to 2$, exact GELU	1,346
M1	None	$39 \to 64 \to 2$, exact GELU	2,690
M2	None	$39 \to 64 \to 64 \to 2$, exact GELU	6,850
M3	Four-by-four history	Raw backbone and pointwise history encoder; mean/latest pooling; $128 \to 2$ head	7,138
M4	Four-by-four history	Raw backbone and $4 \to 32$ gated recurrent unit; mean/latest pooling; $128 \to 2$ head	10,626
M5	History and horizon	M4 with two-entry, eight-dimensional horizon embedding; $136 \to 2$ head	10,658
M6	History, horizon, five basis values	M5 with first two basis values projected to 16 dimensions; $152 \to 2$ head	10,738
M7	History, horizon, five basis values	M5 with all five basis values projected to 16 dimensions; $152 \to 2$ head	10,786
M8	History, horizon, five basis values	M7 plus bias-free $5 \to 2$ linear basis head	10,796

3.4 Numerical qualification

Qualification distinguishes inference from the recovered reference implementation on a central processing unit (CPU) from comparison with archived outputs produced on a graphics processing unit (GPU). For each element, let a_i be the replayed value and b_i the preserved value. The unchanged historical acceptance rule is

$$|a_i - b_i| \leq 10^{-5} + 10^{-4} |b_i| . \tag{1}$$

The original CPU-output comparison has a separate exact-array-equality requirement. The historical arrays contain 512 validation rows and two output columns (1,024 elements per route). The stored qualification used a 24-row synthetic fixture and the preserved 512-row validation fixture. A tolerance pass means every compared array element satisfied Eq. 1; it does not establish that the original GPU execution was replayed.

Fig. 4 shows the 18 model-repeat routes. The result records show exact output equality between the recovered reference implementation and the original CPU out-

Fig 4. Numerical qualification of 18 model-repeat routes (M0–M8, repeats r0 and r1). (a) Recovered CPU outputs are recorded as exactly equal to the original CPU outputs on every route; comparison with archived GPU arrays is within the unchanged elementwise tolerance for M0–M3 and outside it for both repeats of M4–M8. (b) Recorded maximum absolute difference per route on a logarithmic axis. The two outputs have different units (N/kg and m/s), so the values are not comparable as physical magnitudes. A difference below the absolute tolerance 10^{-5} (dashed line) always passes; the criterion also has a relative term, so the maximum alone does not determine status. (c) Number of failing elements out of 1,024 (512 rows by two outputs) where tabulated (n/t: not tabulated). These are recorded comparisons, not a new GPU replay or an accuracy evaluation.

puts for all 18 routes. Comparisons with archived GPU arrays pass for M0–M3 and fail for both repeats of M4–M8. The maximum absolute discrepancy is retained by route in the supplement. Because the outputs have different physical units, a maximum across the two columns is descriptive of array discrepancy only and must not be compared as a physical effect size. The records do not identify the cause of the GPU-array discrepancies.

3.5 Native CMJ measurement and assessment construction

The native measurement contract defines force and impulse on one supplied vertical force trace $F_z(t)$, body-plus-system mass m, gravity g, concentric onset t_{co}, and takeoff t_{to}. Let $T = t_{\mathrm{to}} - t_{\mathrm{co}}$. Gross mean concentric force per unit mass and gravity-adjusted net impulse per unit mass are

$$Y_F = \frac{1}{mT} \int_{t_{\mathrm{co}}}^{t_{\mathrm{to}}} F_z(t)\,\mathrm{d}t, \tag{2}$$

$$Y_J = \frac{1}{m} \int_{t_{\mathrm{co}}}^{t_{\mathrm{to}}} \left(F_z(t) - mg\right)\,\mathrm{d}t, \tag{3}$$

with units $[Y_F] = \mathrm{N\,kg^{-1}} = \mathrm{m\,s^{-2}}$ and $[Y_J] = \mathrm{N\,s\,kg^{-1}} = \mathrm{m\,s^{-1}}$. Peak force, $\max_t F_z(t)/m$, is a third quantity and is not an outcome in this article. The recorded implementation interpolates force linearly at event boundaries and applies the composite trapezoid rule over the closed concentric interval. Event detection, force calibration, mass authority, and trial validity are separate measurement requirements; the integrator does not infer those inputs.

Each native assessment is the arithmetic mean of exactly three distinct valid trials,

Fig 5. Native measurement definitions and assessment chronology. (a) Analytical vertical force–time curve, not athlete data, generated with $m = 70\,\mathrm{kg}$, $g = 9.80665\,\mathrm{m/s^2}$, $A = 1{,}000\,\mathrm{N}$, $t_p = 0.22\,\mathrm{s}$ and $T = 0.55\,\mathrm{s}$: $F_z = mg + A\sin(\pi t/2t_p)$ for $0 \le t \le t_p$ and $F_z = (mg + A)[1 - (t - t_p)/(T - t_p)]$ for $t_p < t \le T$. Dividing the area under F_z by mT gives gross mean concentric force (14.79 N/kg, black line); dividing the signed area relative to body weight by m gives gravity-adjusted net impulse (2.74 m/s). Peak force (24.09 N/kg) is shown only to distinguish it from the scored mean-force target. (b) The three quantities with their equations, values and units. (c) Native contract (upper track): baseline search interval 24–240 h before the index exposure, from which the two latest valid assessments are selected, an H72 window at 66–78 h and a D7 window at 156–180 h; each assessment is the mean of three valid trials. Lower track: the separate public synthetic H24, H48 and H72 query horizons (nominal). No athlete measurements are plotted.

not the best trial or median. For outcome o, define the baseline set $\mathcal{B}_i$ as the two latest distinct valid assessments in the inclusive interval from 24 to 240 hours before origin i. The baseline and target-relative innovation are

$$\bar{Y}_{i,0,o} = \frac{1}{2} \sum_{q \in \mathcal{B}_i} A_{q,o}, \tag{4}$$

$$\Delta Y_{i,h,o} = Y_{i,h,o} - \bar{Y}_{i,0,o}, \tag{5}$$

with $|\mathcal{B}_i| = 2$, where $A_{q,o}$ is the three-trial assessment mean and h is the forecast horizon. The native target windows are 66–78 hours for H72 and 156–180 hours for D7. These are native measurement and evaluation contracts, not an empirical evaluation performed for this article.

Fig. 5 defines the three native quantities that are easily confused and places the native contract on a scaled time axis. Gross mean concentric force (Eq. 2), peak force and gravity-adjusted net impulse (Eq. 3) differ in definition and units: the first is a time average of the force trace, the second is its maximum, and the third is the integral of force above body weight. Panel a is an analytical illustration rather than athlete data; it shows that the gross mean (14.79 N/kg) lies well below the peak (24.09 N/kg) and that net impulse (2.74 m/s) answers a different measurement question. The scored synthetic force outcome described below is the first of these, not the second. Panel c separates the native baseline, H72 and D7 windows from the public synthetic H24, H48 and H72 queries. The native evaluation has four physical-unit cells; the synthetic experiment has six normalized-error cells. They are not combined.

3.6 Native evaluation and inferential rules

The native analysis contract reports four separate cells: force in $\mathrm{N\,kg^{-1}}$ and net impulse in $\mathrm{m\,s^{-1}}$, each at H72 and D7. Within a cell, query-weighted root-mean-square error (RMSE), mean absolute error (MAE), and signed bias are computed from prediction-minus-observation errors. Outcomes are not averaged across units. Repeats r0 and r1 are reported separately; the family summary is the arithmetic mean of the two repeat-level metrics, not an ensemble of their predictions.

For native uncertainty, the locked procedure resamples participant groups with replacement at least 10,000 times. A sampled participant contributes all of that participant's queries, origins, horizons, targets, candidates, repeats, and baselines. Paired comparisons share the same sampled participants. The resulting 95% percentile intervals are marginal, not simultaneous or predictive. The one-sided sign-flip test uses each independent participant's sum of zero-baseline squared error minus the mean squared error of the two model repeats. Its null requires independent clusters and a symmetric distribution of cluster loss differences. For at most 16 participants all 2^n sign allocations are enumerated; larger samples use a recorded sampled allocation scheme and plus-one Monte Carlo correction. Holm adjustment covers the four locked primary native cells.

No native athlete outcome analysis was performed here. The stored engineering rehearsal uses invented values only. With six strictly positive simulated participant improvements, the smallest exact one-sided probability is $1/64 = 0.015625$; across four primary cells its Holm-adjusted value is $4/64 = 0.0625$. This rehearsal correctly cannot meet a 0.05 family-wise threshold and is not evidence about athletes or any recovered checkpoint.

3.7 Public synthetic benchmark design reported with ADEMP

3.7.1 Aims

The existing public fixed-mode study provides a descriptive comparison of four pre-specified reference models for baseline-relative force and net-impulse innovations. Its target is performance on fixed generated test roots from the registered synthetic world. It does not evaluate the historical neural model families or athlete outcomes.

3.7.2 Data-generating mechanisms

The public synthetic data are generated from a fixed-mode discrepancy world. Training contains 96 camps, 24,000 episodes, and 72,000 horizon rows. Development contains 16 camps, 4,000 episodes, and 12,000 rows. Each of three fixed evaluation roots uses a separate 16-camp, 4,000-episode, 12,000-row public-validation stream from the same generator. Each camp contributes 250 episodes; each episode contributes H24, H48, and H72 queries that remain together. Participant, episode, query, and camp identities are checked for split separation. Group keys, labels, future target-time fields, and oracle states are excluded from predictors. The roots are fixed generator outputs, not independent human populations.

3.7.3 Estimand and methods

For each fixed evaluation root, the study target is the descriptive mean NRMSE over its six outcome-horizon cells. The four reference models are zero innovation, a training-only outcome-by-horizon mean, cellwise ridge regression, and a structured comparator. The public predictor schema has 82 source fields; one-hot horizon encoding produces 84 numeric columns. Numeric preprocessing uses training-only means and population standard deviations, with zero scales set to one. The structured comparator uses public exposure primitives, observed context, observed prior innovations, and fixed 24-hour and 84-hour mathematical modes; it excludes hidden effects and oracle states.

Ridge and structured regularization were selected on development six-cell mean NRMSE from the frozen grid $\{0.01, 0.1, 1, 10, 100\}$, with lower alpha breaking ties. All full-data selections chose the lower grid boundary, $\alpha = 0.01$. This is a property of the existing study, not a reason to adjust its test results. The nested sample-size diagnostic uses 24, 48, and 96 camps from one recorded seeded permutation and holds the full-data-selected hyperparameter fixed. The force outcome is the baseline-relative change in gross mean concentric force per unit mass, in $\mathrm{N\,kg^{-1}}$: for each assessment, three simulated concentric-force trials are averaged, and the target is the criterion-assessment mean minus the baseline-assessment mean. It is not peak force and not net impulse. The registered score-cell identifier `relative_peak_mean_force_innovation` is a legacy name retained in the frozen results; it maps to the generated target `label.relative_mean_concentric_force_innovation`. This mapping is separate from the native H72/D7 measurement contract.

3.7.4 Performance measures

For model m, outcome-horizon cell c, and n_c rows, the registered normalized root-mean-square error (NRMSE) is calculated as follows:

$$\mathrm{RMSE}_{m,c} = \sqrt{\frac{1}{n_c}\sum_{j=1}^{n_c}(\hat{y}_{m,c,j} - y_{c,j})^2}, \tag{6}$$

$$\mathrm{NRMSE}_{m,c} = \frac{\mathrm{RMSE}_{m,c}}{\mathrm{SD}_{\mathrm{pop}}(y_c)}, \tag{7}$$

$$\mathrm{SD}_{\mathrm{pop}}(y_c) = \sqrt{\frac{1}{n_c}\sum_{j=1}^{n_c}(y_{c,j} - \bar{y}_c)^2}, \tag{8}$$

$$S_m = \frac{1}{6}\sum_{c=1}^{6}\mathrm{NRMSE}_{m,c}\,. \tag{9}$$

The six cells are formed by three horizons and two outcomes. Each cell is normalized separately before equal-weight averaging, so the score is dimensionless and does not pool force and impulse in their raw units. Every registered cell has at least two

rows and nonzero population standard deviation. $1 - S_m$ is a descriptive research-progress transformation in the stored result summary; it is not a recovery score or a clinical threshold. This article reports existing study results and does not repeat training, model selection, inference, or scoring.

3.8 Statistical summaries

The reported root-level variation is the sample standard deviation across the three evaluation roots. It describes those fixed generated roots; it is not a confidence interval for a real athlete population. The existing result record also contains 5,000 paired whole-camp bootstrap replicates per root, using NumPy PCG64 with seed 49,505,000. Each draw resamples 16 camps, retaining all 250 episodes and three horizon queries for each selected camp. The bootstrap is paired between each model and zero innovation. Percentile endpoints are copied from the stored summary; no new bootstrap is computed.

The stored bootstrap intervals support within-root model-versus-zero comparisons for ridge and the structured comparator. They do not provide a ridge-versus-structured interval. Separate marginal intervals cannot be combined to manufacture that comparison.

4 Results

4.1 RQ1 — Historical reconstructability

The archive supports eight distinct formulation identities and nine historical neural architecture families, but it does not support a complete historical replay. Five direct parent links are recorded and three identities retain unresolved direct ancestry (Fig. 1). The rich-history formulation remains separate from the preseason and episode chains. The threshold formulation is proposed, while fixed-mode discrepancy is an active candidate; neither status is inferred from the synthetic reference-model ranking. Public files from the software release support the formulation and architecture descriptions. Original fitted checkpoints, fitted preprocessing values, and route-level execution ev-

idence remain restricted; the exact CPU and archived-GPU qualification is reported from a privately recorded summary.

Nine neural model families and 18 repeats were represented in the architecture qualification record. Parameter counts range from 1,346 in M0 to 10,796 in M8 (Table 1). The sequence of model changes includes feedforward depth and width, explicit history encoding, horizon conditioning, basis projection, and an additive linear basis head. These recovered topology differences do not by themselves establish which architecture predicts native H72/D7 outcomes better.

4.2 RQ2 — Numerical reproducibility

Exact CPU-output equality between the recovered reference implementation and original outputs is recorded for all 18/18 model-repeat routes. Under the unchanged tolerance, archived GPU-array comparisons pass for 8/18 routes (M0–M3) and fail for 10/18 (both repeats of M4–M8; Fig. 4). The two M8 routes have recorded maximum absolute discrepancies of 2.21×10^{-4} and 3.04×10^{-4}, with 98 and 145 failing elements out of 1,024, respectively. Across the failed families, recorded failure counts range from 88 to 443 elements per route. An external reader can inspect the public summary and arithmetic checks, but cannot independently rerun the original checkpoint inference from this package.

These findings qualify source-level CPU inference and identify a negative historical-runtime comparison. They do not establish a successful original GPU replay or determine whether precision mode, kernel behavior, or another runtime factor caused the discrepancies. The saved arrays' two output columns have different physical units; their pooled maximum absolute error is not a meaningful physical comparison across routes. No saved-array failure is hidden by changing the threshold.

4.3 RQ3 — Scientific comparability

The recorded six-cell results appear in Fig. 6 and Table 2.

The four reference models compare force and impulse at H24, H48, and H72. The

Fig 6. Existing public fixed-mode synthetic benchmark across six outcome-horizon cells. Each panel shows, for the zero-innovation baseline, training-mean baseline, ridge regression and structured comparator, the three fixed evaluation roots (A circle, B square, C triangle; offset horizontally only to separate marks) and the stored mean (black bar, value printed). Rows are H24, H48 and H72; columns are mean concentric-force innovation (N/kg target) and net-impulse innovation (m/s target). The force cells retain the legacy registered identifier `relative_peak_mean_force_innovation`, but the target is mean, not peak, concentric force. The vertical axis is cellwise NRMSE (dimensionless) on a common scale; the dashed line marks NRMSE = 1. Each root contains 16 camps and 4,000 rows per cell. Points and means describe three fixed same-generator roots, not confidence intervals or human cohorts.

Table 2. Six-cell public synthetic benchmark NRMSE. For each model, the first column is the stored mean across three same-generator evaluation roots and the second is the sample standard deviation across those roots, rounded to four decimal places. Full stored values are retained in the accompanying numerical table. NRMSE is dimensionless because each outcome-horizon RMSE is divided by that cell's population standard deviation before averaging. Force targets (mean concentric-force innovation, N/kg) and impulse targets (m/s) are reported in separate rows and never pooled in physical units.

	Zero innovation		Training mean		Ridge		Structured	
Cell	Mean	SD	Mean	SD	Mean	SD	Mean	SD
H24 force (N/kg)	3.0335	0.0220	1.0477	0.0070	0.6244	0.0101	0.6575	0.0114
H24 impulse (m/s)	2.7618	0.0054	1.0451	0.0085	0.6522	0.0017	0.6960	0.0021
H48 force (N/kg)	2.5420	0.0139	1.0395	0.0095	0.7319	0.0096	0.7610	0.0128
H48 impulse (m/s)	2.2644	0.0369	1.0318	0.0080	0.7598	0.0070	0.8046	0.0077
H72 force (N/kg)	2.1090	0.0072	1.0274	0.0069	0.8157	0.0057	0.8429	0.0075
H72 impulse (m/s)	1.8732	0.0255	1.0248	0.0052	0.8336	0.0037	0.8779	0.0019

three roots share one generator; they are not human cohorts. This is evidence about the registered synthetic task, not an accuracy evaluation of the historical neural model families or validation of their native H72/D7 athlete predictions.

Table 2 reports the stored mean and root sample standard deviation for each outcome-horizon cell. These values preserve the recorded cell aggregates; no prediction or model score was recomputed for this article.

The stored six-cell mean scores were 0.736270 for ridge, 0.773316 for the structured comparator, 1.036038 for the training-mean baseline, and 2.430659 for zero innovation. Root sample standard deviations were 0.003349, 0.005490, 0.007308, and 0.008127, respectively (Table 3). Ridge had the lowest descriptive mean error in this generator. No inferential contrast between ridge and structured was recorded; this ranking is descriptive.

Fig. 7 organizes the same stored means by model and cell. Ridge had the lowest mean NRMSE in each of the six cells, and the structured comparator exceeded ridge in all 18 root-cell comparisons by 0.025 to 0.047 NRMSE units (panel b; differences

Fig 7. Model-by-cell error heatmap. (a) Stored mean NRMSE across the three evaluation roots for six outcome-horizon cells (rows) and four reference models (columns); the bottom row is the equal-weight six-cell mean. Lower is better. Colour is centred at NRMSE = 1, the value of a cell-mean predictor: blue is below 1 and red above 1. Outlined cells are lowest in their row. (b) Descriptive structured-minus-ridge NRMSE for each root (A circle, B square, C triangle); positive values mean ridge was lower. Ridge was lower in all 18 root-cell comparisons (0.025 to 0.047). No interval or test is implied, and no ridge-versus-structured confidence interval was recorded.

Table 3. Overall public synthetic benchmark score: stored mean six-cell NRMSE across three evaluation roots and the sample standard deviation (SD) of those three root scores. This is descriptive root variation, not a confidence interval.

Reference model	Mean NRMSE	Root SD
Zero innovation	2.430659	0.008127
Training mean	1.036038	0.007308
Ridge	0.736270	0.003349
Structured	0.773316	0.005490

Fig 8. Recorded camp-bootstrap uncertainty for ridge regression and the structured comparator within each of three synthetic evaluation roots. (a) Stored 95% percentile intervals for each model's mean six-cell NRMSE; markers show the root estimate. (b) Paired intervals for model-minus-zero-innovation NRMSE; values below zero favour the named model over the zero-innovation baseline. Each root contributes 16 camps, with all 250 episodes and three horizon queries per selected camp retained together, in 5,000 paired resamples (PCG64 seed 49,505,000). Intervals are conditional on a root and are not intervals across roots. No ridge-versus-structured interval was recorded.

of recorded values, with no interval or test). Cell errors for ridge and the structured comparator fall below the NRMSE = 1 cell-mean reference in every cell, whereas the training-mean baseline sits at about 1 by construction and zero innovation exceeds 1 throughout.

4.4 Recorded uncertainty, ablations, and sample efficiency

All six stored within-root model-versus-zero percentile intervals for ridge and structured have negative paired model-minus-zero endpoints. They support a reduction in NRMSE relative to the zero baseline within the corresponding synthetic root under the recorded camp-resampling scheme (Fig. 8). They do not support an uncertainty interval for ridge versus structured or for a new root drawn from a different generator.

4.5 Ablations and sample efficiency

The recorded ridge ablation scores were 0.819681 without prior history, 0.867735 without current exposure, and 0.736346 without participant context, compared with

Fig 9. Recorded ablation and nested-camp comparisons. (a) Evaluation-root values (A circle, B square, C triangle) and means for full ridge regression and variants without prior history, current exposure or participant context, plus the structured comparator without prior history; parentheses give the change in the mean from full ridge. (b) Root-specific and mean scores for ridge regression (solid) and the structured comparator (dashed) trained on nested subsets of 24, 48 or 96 camps, with the regularization parameter selected on the full training and development data held fixed. All values are dimensionless six-cell NRMSE within the same synthetic generator. No intervals are shown; these comparisons do not isolate causal feature effects or estimate athlete-level sample efficiency.

Fig 10. Historical reproducibility gaps. (a) Difference in horizon-row count (public reconstruction minus historical record) for the rich-history camp materialization: training 15,134 versus 15,182 rows (-48); validation 3,799 versus 3,795 rows ($+4$). These are membership differences in the preserved splits, not response effects or uncertainty intervals. (b) Evidence boundary: the class of evidence that supports each result as described in this article, from public source and contracts, through results regenerable from public receipts and privately qualified results available only as public summaries, to what is not established (historical split identity, the cause of the GPU-array differences, public reruns of the private checkpoints, and any athlete predictive validity, transportability or clinical use).

0.736270 for full ridge. Omitting context changes the mean by approximately 0.000076. The separately specified structured model without history scores 0.820342. With 24, 48, and 96 training camps, nested ridge scores were 0.736889, 0.736602, and 0.736270; the structured-model scores were 0.773624, 0.773621, and 0.773316 (Fig. 9). These generator-conditional diagnostics do not identify causal feature importance or human sample efficiency.

4.6 Historical reproduction limits

The public reproduction record remains partial for the historical program. In the rich-history reconstruction, the publicly reconstructed training materialization has 15,134 horizon rows versus 15,182 in the historical record. The validation materialization has 3,799 versus 3,795. These are documented differences in horizon membership; they are not a newly discovered response effect. Matching counts in other formulations would not prove historical split identity.

Fig. 10 quantifies these sample-membership differences as publicly reconstructed minus historical row counts. The M0–M8 qualification is not an empirical athlete performance result.

5 Discussion

This reconstruction illustrates why reproducibility should be reported as a set of qualified operations rather than one binary label. The recorded CPU results from the reference implementation agree exactly across all 18 neural model routes, yet ten comparisons with archived GPU arrays fail the original tolerance. A single "reproduced" label would obscure that distinction. The evidence supports recovered-source inference under the recorded CPU comparison and a negative historical-runtime comparison; it does not show a successful replay of the original graphics-processor execution.

The synthetic comparison is also bounded. Ridge has lower mean six-cell NRMSE than the structured reference in the three supplied roots, while the structured result remains substantially below the two simpler baselines. This is a useful observation about the registered generator, predictors, alpha selection, and aggregation rule. It is not a universal model ranking. Every root comes from the same simulation world, and all six outcomes are normalized separately before aggregation. The recorded bootstrap compares each model with zero innovation but does not test ridge against structured.

The ablation results reinforce this scope. Removing current exposure or history increases error in this generator, whereas removing participant context changes the full-ridge score very little. Because variants were fitted and selected under the existing study protocol, these contrasts describe model configurations under that protocol; they do not identify causal athlete mechanisms. Likewise, the small difference across nested camp sizes cannot be generalized to a real monitoring program, particularly because each synthetic camp contains many generated episodes and the model's regularization value was selected on the full training/development data then held fixed.

Measurement contracts place a separate boundary on interpretation. Correct trapezoidal integration requires correct force calibration, system mass, event boundaries, and trial validity. A stable arithmetic formula does not prove that an empirical event detector selected the intended phase or that a two-assessment baseline represents physiological recovery. The native H72/D7 metrics are in physical units and are not interchangeable with the public benchmark's H24/H48/H72 cellwise normalized errors.

The contribution is a case study of one historical program and an engineering account of how its evidence can be separated. Stable source identities, explicit route and measurement contracts, a frozen tolerance, arithmetic checks of the stored results, and a public summary make the supported results inspectable. This application draws on established reporting, tracking, provenance, and artifact-review practices; it does not introduce a new general framework, experimentally compare those systems, or demonstrate that the approach generalizes to other archives [2, 4–8, 12]. The technical supplement (S1 Text) contains the formulation, route, cell, and bootstrap tables.

6 Limitations and threats to validity

First, the original model weights, configurations, and restricted records are not part of the public data package. The private artifact associations and exact CPU-output equality are preserved as qualification evidence, but a reader with only the released package cannot independently reproduce those original inference outputs. This access boundary is a limitation of independent auditability, not permission to disclose unlicensed artifacts.

Second, archived GPU-output discrepancies are reported under a fixed tolerance, but the exact historical runtime cause is unresolved. Maximum absolute errors pool two output columns with different units, so they must not be interpreted as a common physical magnitude. No GPU replay or new checkpoint inference was performed for this article.

Third, the public benchmark has only three evaluation roots from one synthetic generator and 16 camps per root. The root sample standard deviation is descriptive. The camp bootstrap is conditional on each root and its resampling design. Neither estimates transportability to athlete populations, and the stored bootstrap does not compare ridge directly with the structured comparator.

Fourth, the alpha grid selection reached its lower boundary for all full-data model families. The recorded test results are retained without retuning, but performance may

depend on regularization values outside the frozen grid. The nested sample-size analysis holds the full-data-selected alpha fixed and should not be read as a complete study of training-data efficiency. For any empirical prospective setting, the temporal authority of the future-plan snapshots used by the original schedule basis would also need to be established.

Fifth, no independent athlete cohort, prospective assessment protocol, clinical-use evaluation, or production-monitoring study is included. No human predictive validity, causal response mechanism, out-of-distribution performance, clinical utility, or individual recovery classification is established. The eight historical formulations have partial reproduction status, and the rich-history row-count discrepancies remain unresolved.

7 Reproducibility and availability

The public research software is available under Apache-2.0 at https://github.com/Litju/CMJRecoveryDynamicsBench/releases/tag/v1.1.0. Its immutable source identity is commit `401f87aad1e4ef4108011fa0c1d769a24f1ad329`. The public data package includes the synthetic-study protocol and results, versioned source files, summarized route findings, calculation scripts, figure data, and SHA-256 checksums. An external reader can verify source-file hashes, reproduce the figures, and check the reported synthetic-study arithmetic using those materials. The public data package does not contain the original fitted checkpoints, fitted preprocessing values, private inference arrays, or restricted observation and source-reconciliation records. It therefore cannot independently reproduce the recorded original-model qualification. No approved controlled-access route for those restricted artifacts is established here, and access is not offered upon request. The public synthetic results do not validate the restricted historical model-route findings. The accompanying availability assessment maps each main claim to the public materials and identifies what cannot be independently reproduced.

8 Conclusions

This case study answers three bounded questions. Eight historical formulation identities and nine original neural model families can be described from the preserved archive, but ancestry, fitted state, rights, and rich-history membership remain partly unresolved. The privately recorded numerical qualification reports exact CPU-output equality for 18 routes and tolerance failures against archived GPU outputs for both repeats of M4–M8; it does not establish historical GPU replay and cannot be independently rerun from the public data package. The public synthetic benchmark reports lower descriptive six-cell NRMSE for ridge than for the structured comparator across three roots from one generator, but these are not accuracies of the historical model families or athlete validation. The contribution is the evidence-bounded reconstruction of this program, not a universal provenance method or a claim of predictive validity.

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Data availability. The v1.1.0 source release, public synthetic-study protocol, results,

source files, and figure data are available at the release URL and in the accompanying data package. Original fitted checkpoints, fitted preprocessing values, private inference arrays, and restricted observation and source-reconciliation records are not included. This package does not establish a controlled-access mechanism for those materials; they are not offered upon request. Public materials permit figure regeneration and verification of the published result arithmetic, but not independent reproduction of the original checkpoint qualification. See the accompanying claim-level data and software availability assessment. Journal-specific eligibility remains unresolved.

Generative AI and AI-assisted technologies. AI assistance supported official-guideline research, manuscript restructuring and drafting, plotting and document-generation code, and first-pass artifact review. Benchmark and qualification values come from pre-existing scientific records; conventional Python code generates the figures, including the equation-based illustration in Fig. 5a. No generative-image model or new empirical benchmark result was used. The human authors have not yet reviewed or approved this draft and must verify its content, sources, and all declarations before submission. No AI system is an author.

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Supporting information captions. S1 Text. Technical supplement with formulation, architecture, route, synthetic-results, bootstrap, ablation, protocol, and native-measurement tables. Public summaries only; no restricted artifacts are included.